



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2025-26
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : PHY F212
Course Title : Electromagnetic Theory 1
Instructor-in-Charge : Rakesh Choubisa
Instructor(s) : Rakesh Choubisa and Sandipan Dutta
Tutorial/Practical Instructors: Rakesh Choubisa and Sandipan Dutta

1. Course Description: Electromagnetic Theory I is the first of two courses in Electromagnetic Theory for Physics students and is also required for Chemistry students. This course covers topics in electricity, magnetism, and electromagnetic waves.

2. Scope and Objective of the Course:

- Learn the mathematical tools needed to solve problems in electromagnetism
- Understand the concept of electric field
- Solve electrostatic problems using the concept of potential
- Understand how electric field is modified in materials
- Understand the concept of magnetic field
- Understand how magnetic field is modified in materials
- Learn about magnetic induction and the basics of electrodynamics
- Introduction to Maxwell's equations and electromagnetic waves

3. Text Books: “Introduction to Electrodynamics”, David J. Griffiths, Third Edition, Pearson Education Inc., 1999

4. Reference Books: “Berkeley Physics Course Vol. 2, Electricity & Magnetism”, Edward M Purcell, Special Indian Edition: Tata McGraw-Hill Publishing Company Limited

5. Course Plan:

Lecture No.	Learning Objectives	Topics to be covered	Chapter/Section
1-4	Vector Algebra	Gradient, Divergence and Curl, Line, Surface and Volume Integrals, Curvilinear co-ordinates, Dirac Delta Function	1.2-1.6
5-9	Electrostatics	Divergence and Curl of electrostatic fields, Electric potential, Work and Energy in Electrostatics	2.1-2.4
10-12	Conductors	Conductors, Induced charges, Capacitors	2.5
13-15	Special techniques	Laplace's Equation, Separation of Variables, Multipole expansion.	3.1, 3.3, 3.4
16-19	Electric Fields in Matter	Polarization, bound charges, electric displacement, and Dielectrics	4.1-4.4
20-25	Magnetostatics	Lorentz force law, Biot-Savart law, Ampere's law, Magnetic Vector potential	5.1-5.4



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26-30	Magnetic fields in Matter	Magnetization, the field of a magnetized object, Ampere's law in magnetized materials, Magnetic susceptibility and permeability, Ferromagnetism	6.1-6.4
31-35	Electrodynamics	Electromotive force, Ohm's law, Electromagnetic Induction, Faraday's law	7.1-7.2
38-40	Maxwell's Equations	Maxwell's equations, Boundary conditions	7.3

6. Evaluation Scheme:

<i>EC No.</i>	<i>Evaluation Component</i>	<i>Duration</i>	<i>Weightage (%)</i>	<i>Date, Time & Venue</i>	<i>Remarks</i>
1	Mid. Semester Test	90 Minutes	30	9/10/26, AN2	Closed Book
2	Tutorial tests	30 Minutes each	21	<i>To be announced in the class</i>	Closed Book (announced): Out of four tutorial tests, the best three would be considered.
3	Assignment	To be announced	9	<i>To be announced in the class</i>	Open Book.
4	Comprehensive Examination	3 Hours	40	14/12/26, AN	Closed/Open Book

7. Chamber Consultation Hour: To be announced in the respective tutorials and lecture classes.

8. Notices: All notices will be posted on Nalanda/Google Classroom and also be announced during class.

9. Make-up Policy: Makeup will be given only for genuine cases (accompanied by proof) for the Mid-semester or comprehensive examination only. The decision of the instructors will be final regarding the approval of makeup exams for both mid-semester and comprehensive examinations.

11. Criterion for NC: If you do not take part in a complete evaluation component or the post-comprehensive marks is less than 10% of the total coursework, NC will be awarded.

12. If your attendance is below 50% of the classes conducted, your marks for the assignment will not be considered.

Instructor-in-charge
Course No. PHY F212